

Insidor: detection to coin

A Solana web app that finds viral moments before crypto traders do, lets you buy the coin that exists or create the first one, and pays a share of fees to whoever made the original post.

This document is the result of two independent research passes: an adversarial fact-check of 120 extracted claims against 24 primary sources, and an eight-discipline engineering design that was then reviewed for contradictions. **Both passes found showstoppers.** They are in section 0, not buried.

24 sources 25 claims verified · 9 killed 8 design layers 12 contradictions found

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The verdict

Three findings survived adversarial verification and each one independently threatens the product as specified. Read these before anything else.

SHOWSTOPPER 1 · THE HEADLINE MECHANIC HAS NO TRUSTLESS IMPLEMENTATION

Routing a fee share to a poster identified only by a social handle, who never consented and has no wallet, **cannot currently be done trustlessly on Solana.**

- **Bags** exposes `GET /token-launch/fee-share/wallet/v2` which resolves a handle to a wallet [9] — but its documentation is *silent* on what happens for a handle that has never onboarded. That single undocumented behaviour decides the feature.
- **PumpPortal**'s `collectCreatorFee` harvests only to the caller's own wallet, and on the `pump.fun` pool it sweeps all tokens at once [10].
- **Token-2022 transfer fees** are withheld on the *destination* account by deliberate design to preserve parallelism, and require a project-held withdraw authority to collect [11].

Every verified path is custodial. That converts your marquee feature from a smart-contract problem into an **unclaimed-funds and money-transmission problem**. It is a legal question with an engineering consequence, and it must be answered before Phase 5.

SHOWSTOPPER 2 · PLATFORM POLICY MAY FORBID THE MECHANIC OUTRIGHT

On 15 January 2026, X revoked API access for "InfoFi" applications — apps attaching financial rewards to posting [4][20]. Insidor routes money to posters. The distinction we are relying on is that **the payout triggers on trading volume, never on posting**, which is structurally a royalty rather than a bounty on amplification.

That distinction is **ours, not X's**. The verification pass rated this finding *low confidence* precisely because nobody has read the mechanic against the actual Developer Agreement text [19] rather than secondhand reporting. Get counsel to read it before building on X.

SHOWSTOPPER 3 · TIKTOK IS NOT AVAILABLE, ON TWO INDEPENDENT GROUNDS

A for-profit entity is **categorically ineligible** for TikTok's Research API under its own eligibility criteria and binding Research Tools ToS [2]. Separately, it runs roughly 48 hours behind — useless for a product measured in minutes. The Commercial Content API is regulator-gated; the Display API carries no discovery data; Instagram's hashtag search is capped at 30 hashtags per user per rolling 7 days.

There is no lawful first-party discovery API for TikTok or Instagram. The architecture below therefore treats X as the sensor for every platform and non-X sources as hydration targets only.

AND ONE STRUCTURAL FINDING FROM THE DESIGN REVIEW

Eight independent specialists designed the *create* path in exhaustive detail. **Not one of them designed story-to-existing-coin matching** — mapping a detected story to a mint that already exists. That is the buy path, it is the majority of trade volume, and it is the thing the product promises first. It is now assigned explicitly in Phase 4.

01

What the product is

Insidor watches social media for things going viral that have nothing to do with crypto, surfaces them before crypto traders find them, and lets a user either buy the coin that already exists or create the first one. When creating, the coin's name, ticker and image are generated from the source post. Creator fees split three ways: **50% creator, 40% original poster, 10% Insidor**.

The one number

Lead time — minutes between Insidor detecting a story and the first crypto mention of it. It is the value proposition, the share hook, and the only claim that must be provable to a sceptic without trusting us.

The object model

Eight objects. **Story and Coin are separate**: one Story has 0..n Coins, a Coin belongs to exactly one Story. Everything else follows from that split.

OBJECT	IS	KEY RELATIONSHIPS
Story	A viral moment	1 Post, 1 Poster, 0..n Coins
Post	The source item	Its permalink is the provenance proof
Poster	Who published it	Owed 40% whether or not they know
Coin	An SPL mint	$n \rightarrow 1$ Story
Follow	A watch	Targets a Story or a Coin
Holding	A position	1 Coin, 1 Story
Launch	A coin you made	1 Coin
Payout	Fees owed	3 legs per Coin

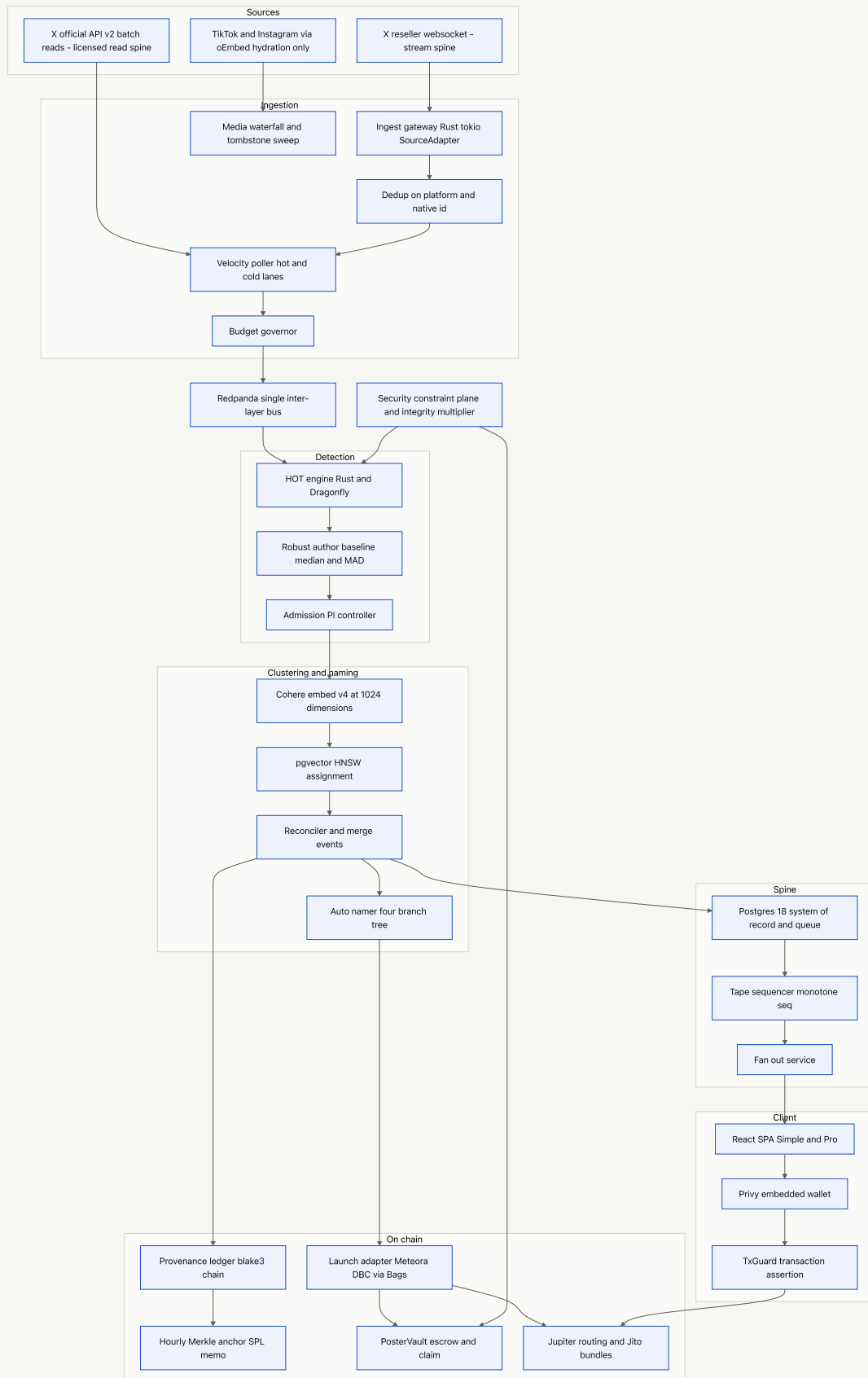
02

System context

Eight layers. Conflicts found in review are marked on the diagram rather than hidden.

CONTAINER DIAGRAM

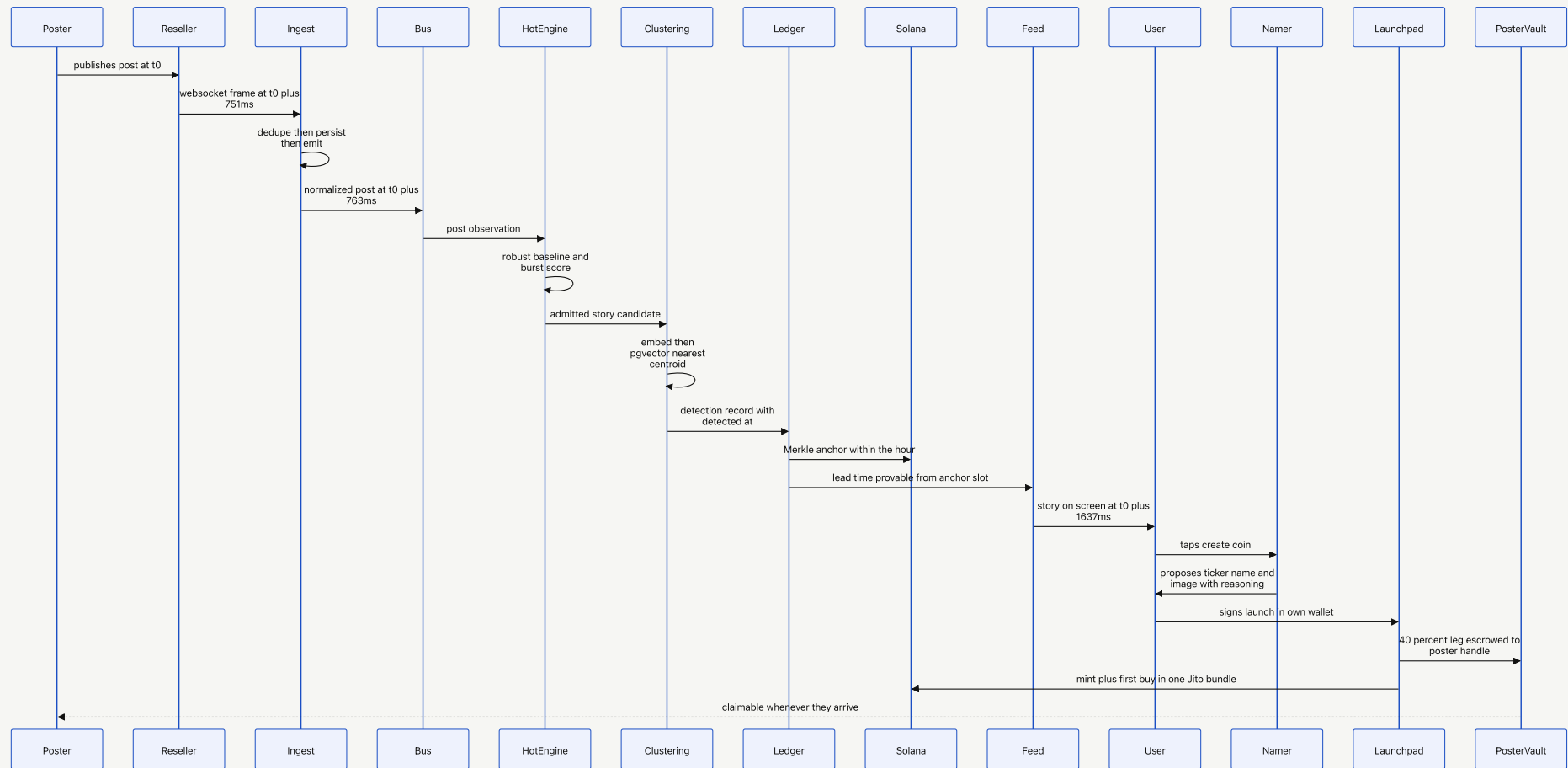




The critical path

Post published to coin created, with the real measured timings.

SEQUENCE · DETECTION TO COIN



Ingestion

Rust + tokio gateway, Redpanda bus, Postgres + TimescaleDB, Cloudflare R2 archive.

THE FINDING THAT INVERTS THE OBVIOUS DESIGN

The instinct is "official X stream as the licensed spine, reseller as the volume lane." **The roles must invert.** Filtered stream is not sold on pay-per-use — it starts at Pro (~\$5,000/mo), which at 1k DAU is 3× the entire ingestion budget. But `GET /2/tweets` batch reads *are* pay-per-use at \$0.005/read with same-UTC-day dedup.

So: **reseller websocket is the stream spine, official X is the read spine.** Official filtered stream turns on at Pro only above ~30k DAU — a config flag, not a rewrite.

READ-ROUTING BREAK-EVEN

$$n* = C_{\text{official_per_post_per_utc_day}} / C_{\text{reseller_per_read}} = \$0.005 / \$0.00015 = 33.3 \text{ polls per post per UTC day}$$

Poll a post more than 33×/day → official (flat, dedup makes it free after the first). Fewer → reseller. The hot lane runs 96 polls/day, the cold lane 12. **This one number governs roughly 70% of the ingestion bill.**

ASSUMPTION THAT MUST BE KILLED IN WEEK 1 — COST \$0.02

The entire routing model assumes X's same-UTC-day dedup is a **billing** dedup, not a **response cache**. If it caches responses, dense re-polling returns stale `public_metrics` and **the velocity signal dies entirely**.

Test: read one high-velocity post twice 90s apart in the same UTC day; assert `like_count` differs; then `GET /2/usage/tweets` and assert usage delta is 1, not 2.

X is the sensor for every platform

Since no lawful discovery API exists for TikTok or Instagram, X rules match outbound links — `url:"tiktok.com/@"`, `url:"instagram.com/reel"` — and the URL is then hydrated through each platform's own public oEmbed. `Post.platform` is `tiktok` with a `VIA_X_LINK` provenance bit set. Anything viral enough to be worth a coin gets cross-posted to X within minutes.

Two hard constraints on the timing story

- **~20s rule activation floor.** Rules mutate live without reconnecting [1], but "most rule additions take effect about 20 seconds or less." You cannot react to a brand-new term — you must already be listening.
- **Semantic matching is Enterprise-only.** The `embedding:` operator matches by meaning rather than keyword, but requires Enterprise plus an Embedding tier entitlement [1].

Baseline rule design must assume keyword and structural matching only.

Media waterfall

Five tiers, fixed card height across all five, so there is never a broken image:

1. `media.url` photo — free, in the stream payload
2. `preview_image_url` for video — free, **no ffmpeg needed**, X generates the still server-side
3. `og:image` of the first outbound link — own fetch, free
4. Image from the highest-scored reply — capped at 5, gated behind $HOT \geq p95$, since at \$0.005/read this is the most expensive operation in the product
5. Deterministic typographic plate keyed on `hash(story_id)` — same story always looks the same, which is itself a provenance property

05

Virality detection

Rust stream workers scoring in-process against Dragonfly; LightGBM lambdarank for coefficient fitting on ClickHouse.

CORRECTION TO SOMETHING STATED EARLIER IN THIS PROJECT

X's 2026 open-sourced ranker [5] gives the **shape** of a scorer — a linear weighted sum of predicted action probabilities, multiplicatively attenuated — but ships **zero numeric coefficients**, and has deleted SimClusters, TwHIN and real-graph entirely. The widely-circulated 2023 weights (favorite 0.5, retweet 1.0, reply 13.5, reply-engaged-by-author 75.0) **cannot be attributed to it**. Verification also killed the claim of a specific 15-action Phoenix probability set.

Consequence: Insidor must fit its own coefficients. The 2023 weights are usable only as a *prior*, never as a citation.

ACTION WEIGHT, DERIVED NOT BORROWED

$w_a = C / p_a$ where p_a = platform base rate of action a per impression $C = 0.0075$, fixed by anchoring X repost to $w = 1.0$

The insight behind X's published weights is that weight is the inverse of rarity. Deriving `w_a` from our own observed base rates reproduces that logic without depending on numbers we cannot verify, and it generalises across platforms where base rates differ wildly.

WEIGHTED ENGAGEMENT AND RATE

$$E_w(p,t) = \text{SUM}_a [w_a * n_a(p,t)] + (w_{\text{complete}} / D_{\text{sec}}) * \text{total_watch_seconds}(p,t) \\ R(p,t) = E_w(p,t) / \max(I(p,t), 1)$$

`n_a` = cumulative count of action `a`; `D_sec` = video duration; `I` = impressions (impression_count on X, play_count on TikTok). The watch-time term is what makes TikTok comparable to X at all.

ROBUST SCALE – WHY MEDIAN AND MAD, NOT MEAN AND STDDEV

$$\text{sigma_hat} = \max(1.4826 * \text{MAD}, 0.5 * (Q90 - Q50) / 1.2816, \text{sigma_floor_cohort}) \\ \text{MAD} = \text{median}(|x - \text{median}(x)|)$$

1.4826 is the Gaussian consistency constant; 1.2816 is the 90th-percentile *z*. A viral post is the outlier you are measuring — using mean and standard deviation lets the event you are detecting corrupt its own baseline. This is Twitter's own Seasonal Hybrid ESD reasoning [8], which substitutes median for mean and MAD for stddev.

Burst detection

Kleinberg's automata [7] are fully specified and directly implementable. Two variants matter, and the distinction is the one most implementations get wrong:

- **A*** operates over raw inter-arrival gaps. Good, but it is a batch/windowed dynamic program — not a streaming detector.
- **B*** operates over *windowed counts*, which is the realistic ingestion shape, and produces a **baseline-normalised burst weight** that ranks competing viral moments against each other. That ranking is exactly what a feed needs.

Verification killed the specific claim that Kleinberg's cost recurrence is "identical in form to Viterbi" — implement from the paper, not from that description.

THE DELIBERATE INVERSION – THE WHOLE PRODUCT RESTS ON IT

X applies a **0.75 multiplier** to out-of-network posts and gates them behind social proof. Insider **inverts it**: small accounts with no second-degree connections get a *boost*, because that is exactly where memecoin alpha lives and exactly what the platform is structurally worst at surfacing.

The review found this specified **four different ways spanning 8×** at 100 followers — the precise band where Sybil farms live. One `0mega`, owned by detection, version-locked to the security purity gate. Everything else is dead code.

06

Clustering and the auto-namer

Cohere `embed-v4.0` at 1024-d Matryoshka, stored as `halfvec(1024)` in Postgres 17 + pgvector 0.8 with a partitioned HNSW index over hot stories only.

The decisive reason for embed-v4 is that it is **natively interleaved text and image in one space** — a TikTok still and an X caption about the same moment land near each other without a separate fusion step. Cost is negligible: **\$16/month at 500k DAU** [16].

STORY ASSIGNMENT SCORE

$\text{score}(p,s) = 0.68 * \cos(e_p, c_s) + 0.14 * J(E_p, E_s)$ Jaccard over extracted entities
+ $0.08 * \text{host_overlap} + 0.06 * \text{media_match} + 0.04 * \exp(-dt_min/240) -$
 $0.20 * \text{lang_mismatch} * (1 - \cos)$

e_p = L2-normalised embedding, c_s = story centroid. Assign if score exceeds threshold, else spawn a provisional story under an advisory lock keyed on the nearest centroid — which kills the concurrent-duplicate-spawn race.

CENTROID UPDATE

$\alpha = \max(0.02, 1/(n+1))$ $c_new = (1-\alpha) * c_old + \alpha * e_p$, then $c_new /= ||c_new||$

The 0.02 floor stops a large story's centroid from freezing, so a story that mutates (a meme that acquires a new catchphrase) can still follow its own drift.

The auto-namer

Four branches in priority order, derived from 36 real cases. It shows its reasoning, because visible reasoning is what makes a generator feel intelligent rather than arbitrary.

BRANCH A proper noun confidence 85–95

Does an animate subject have a name in text, caption or comments?

The ticker IS the name, unmodified. Nazgul -> \$NAZGUL

Five of the six largest animal conversions follow this with zero creativity.

If this hits, STOP.

BRANCH B stressed catchphrase noun 60–80

Take the single most-shouted word, never the whole phrase.

"hawk tuah" -> \$HAWK "chicken jockey" -> \$JOCKEY

BRANCH C deliberate misspelling 45–65

Phonetic, deliberately wrong description of what is in frame.

a dog wearing a hat -> \$WIF (the misspelling is load-bearing)

BRANCH D object noun 40–60

Most distinctive concrete object. Kool-Aid pineapple -> \$JARS

HARD BLOCKS – never proposed

- initialisms when any concrete noun exists

\$MVG sat on a 4-billion-impression event and died while
the word "gorilla" was right there

- numbers unsearchable, collide with everything

- registered brands where the joke is at the brand's expense

- the creator's own real name

Critical input most implementations miss: **top comments and quote-posts**. "dogwifhat" was named by a replier, not the poster. "just a chill guy" was a caption someone else attached. **The crowd usually names the thing before the product does.**

07

On-chain

Launch on Meteora Dynamic Bonding Curve

(`dbcij3LWUppWqq96dh6gJWwBifmcGfLSB5D4DuSmaqN`), reached in Phase 1 through the Bags public API because its Fee Share v2 program already ships handle-to-wallet resolution [9].

THIS IS WHERE SHOWSTOPPER 1 LIVES

Bags takes a **protocol cut before any creator-side split**, so a fee-claimer's basis-points share applies to the creator half only — bounding what you can actually pay a poster at roughly **half the naive figure**. The "40%" in the UI is 40% of the creator leg, not 40% of all fees, and the product copy must say so.

Verification also **killed** the claims that splits must sum to exactly 10,000 bps and that fee-share config is immutable after launch. Do not design against those until re-verified.

Provenance — the one component that cannot be rented

An append-only, ed25519-signed, blake3-chained ledger written **at detection time, before any coin exists**, Merkle-anchored hourly to Solana in a single SPL memo instruction.

LEDGER AND ANCHOR

```
leaf_hash = blake3(0x00 || canonical_cbor(record)) invariant : record[n].prev_hash == leaf_hash(record[n-1]) node(L,R) = blake3(0x01 || L || R) odd level duplicates last proof_len = ceil(log2(n_leaves)) anchor_cost = 5000 * n_sigs + (cu_limit * price_microlamports) / 1e6 = 5000 + (16000 * 25000)/1e6 = 5400 lamports ~ $0.0012 per anchor
```

At hourly anchoring that is **about \$60/year** — and it is the only thing in the system that makes "we saw it first" checkable by a sceptic with no dependency on us. Ship it in Phase 1; every hour it is not running is permanently unrecoverable.

Execution

- **Jito bundles** for create + first buy atomically, tip from `getTipFloor`. A one-click launch off a public feed is a sniper magnet by construction [12].
- **Jupiter** for swap routing with integrator fees and pre-created referral token accounts [14].
- **Helius LaserStream** gRPC for mint and trade events [13], behind an internal stream interface so the vendor is swappable.

Data model and spine

One PostgreSQL 18 primary is the entire spine: system of record, job queue via `SELECT ... FOR UPDATE SKIP LOCKED`, vector store via pgvector 0.8 HNSW, and realtime transport via logical replication consumed by a Go fan-out service.

The decisive reason is that Insidor's write load is set by *admitted stories*, not raw posts — the admission controller holds that at a constant rate by design. That is a Postgres-shaped workload for a long time.

TAPE HASH CHAIN — CLIENTS CAN DETECT GAPS AND TAMPERING

```
h(0) = sha256("insidor.tape.v1") h(n) = sha256( h(n-1) || u64be(seq_n) ||
frame_id_n || u8(kind_n) || subject_id_n || u64be(ts_micros_n) ||
sha256(canonical_json(d_n)) ) prefix-visibility invariant: for any MVCC snapshot S,
visible(S) = { r in tape : r.seq <= W(S) }
```

Holds because exactly one session assigns `seq`. A client that reconnects can prove it missed nothing.

VELOCITY AND ACCELERATION

```
v(p,t) = ( e_score(t) - e_score(t - dt) ) / (dt/60) e-units per minute a(p,t) = (
v(p,t) - v(p, t-dt) ) / (dt/60)
```

`dt` = poller interval at that age. Acceleration — the second derivative — is what catches the ripple rather than the wave, and it is a sortable column in Pro.

Client

React 19 + Vite 6 as a pure client-rendered SPA, with a small Cloudflare Worker serving crawler-only OG HTML for share cards.

No Next.js or RSC: the app is a persistent-WebSocket authenticated terminal where every *byte of state is client state*, so server components buy nothing and cost complexity. The Worker exists solely so a shared link unfurls — and it renders from a **signed lead receipt**, so a shared card can never carry a stale number.

ARRIVALSCHEDULER — WHY CLICK TARGETS NEVER MOVE UNDER A FINGER

```
insertY = (rank - firstVisible)*ROW_H - (scrollTop mod ROW_H) safe = (now -
lastPointerAt > GUARD_TTL_MS) OR (insertY > pointerY + GUARD_RADIUS_PX)
interactionLocked(now) = (now - lastPointerDownAt < 600) OR (now - lastWheelAt <
220) OR (now - lastTouchMoveAt < 220) OR |scrollVelocity| > 40 px/s OR armedTicket
!= null
```

This is the single largest contributor to the chaotic feeling in competing terminals. Inserting above a reading user is the cardinal sin; the scheduler makes it structurally impossible.

Wallets: Privy embedded wallets plus Wallet Standard behind one `WalletPort` interface, so a user who has never held crypto can start, and a power user can bring their own. **Insidor**

never custodies funds — which is exactly why showstopper 1 is dangerous, since the PosterVault escrow is the one place that principle bends.

10

Security and anti-abuse

Postgres as a constraint-enforcement plane — the anti-manipulation rules are UNIQUE constraints and CHECKs, not model outputs.

THE FINDING THAT SAVES AN ML PROJECT

Research shows per-account bot scores **do not correlate well** with coordinated liking behaviour. So structural limits beat classifiers: **one contribution per author per story per 30-minute window, enforced as a database UNIQUE constraint**. $O(1)$, no model, no training set, and it structurally defeats engagement pods. Ship this before any bot classifier.

SUBLINEAR SYBIL DAMPING

$S_{\text{eff}} = \text{SUM over clusters } c \text{ of } (n_c^\alpha)$, $\alpha = 0.35$
1 account $\rightarrow 1.00$
100 accounts $\rightarrow 5.01$
1000 accounts $\rightarrow 11.2$

Buying 1000 accounts buys 11x the influence of one, not 1000x. It makes farming economically pointless rather than technically impossible, which is the only defence that holds.

COORDINATION EDGE — MACHINE-CHECKABLE, NOT VIBES

$E(a,b) = 1$ iff $|S_a \cap S_b| \geq 5$ AND $J_w(a,b) \geq 0.55$ AND $\text{median } |t_a(s) - t_b(s)| \leq 180s$ AND $|S_a \cap S_b| / \min(|S_a|, |S_b|) \geq 0.40$

Two accounts are coordinating if they hit the same 5+ stories, with high weighted Jaccard, within 3 minutes of each other, and that overlap dominates at least one of their histories.

Burstiness (Goh-Barabási) $B = (\sigma_\tau - \mu_\tau) / (\sigma_\tau + \mu_\tau)$ over inter-arrival times separates organic spread (bursty, $B \rightarrow 1$) from scripted engagement (regular, $B \rightarrow -1$).

11

Latency budget

Assembled from every layer's own numbers, **including five stages that no layer counted**.

#	STAGE	OWNER	P50	P99	NOTE
1	Source to reseller edge	vendor	251 ms	~1400 ms	Self-reported; p99 unverified
2	Stream interval quantisation	Ingest	500 ms	1000 ms	Provider default 60s would cost 30,000 ms
3	Normalize, dedupe, persist	Ingest	12 ms	45 ms	Persist before emit, for replay idempotency
4	Bus hop	nobody	15 ms	60 ms	Exists only because two layers picked different buses
5	HOT scoring	Detection	2 ms	12 ms	Only stage that beats its own claim
6	Score emit quantisation	Detection	500 ms	990 ms	1 Hz cap, never counted as latency
7	Embedding call	Clustering	120 ms	600 ms	Unbudgeted hosted network hop
8	kNN assign, centroid update	Clustering	12 ms	90 ms	Sub-4ms claims were extrapolation
9	Integrity multiplier	Security	3 ms	25 ms	Inputs up to 60 min stale
10	Admission threshold	Detection	1 ms	5 ms	Threshold up to 5s stale
11	Publish and fan-out	Backend	20 ms	120 ms	Commit + logical decoding + serialise
12	Network to client	—	40 ms	200 ms	Protocol conflict: SSE vs WebSocket
13	Decode, drip, paint	Client	161 ms	1100 ms	Includes deliberate ArrivalScheduler pacing
Detection to screen			1,637 ms	~3,200 ms	Honest figure

AGAINST THE CLAIMS

The backend layer claimed **491 ms p50** for the whole system — it omitted five of thirteen stages, i.e. five of eight layers. SRE claimed a **9,000 ms SLO** it never decomposed. Neither number describes the system these designs actually build. **1.64 s p50 is the honest figure**, and it is still comfortably inside the product's promise, because the competition is measured in *minutes*.

Cost model

Every layer priced its own dependencies, so the naive sum double-, triple- and quintuple-counts the shared substrate.

LAYER	1K DAU	50K DAU	500K DAU
Ingestion	\$2,027	\$11,071	\$56,620
Detection	\$2,518	\$12,813	\$51,623
ML clustering and namer	\$697	\$2,355	\$4,734
On-chain	\$1,173	\$5,870	\$34,988
Backend spine	\$1,297	\$6,844	\$19,900
Client	\$434	\$2,411	\$10,895
Security	\$4,280	\$17,666	\$85,057
SRE and platform	\$4,970	\$37,070	\$145,680
Naive sum	\$17,396	\$96,100	\$409,497

The X firehose was claimed by **four** layers, Solana RPC by **five**, ClickHouse by four, Redis-family by four across three vendors. After deduplication and one decision per shared component:

RECONCILED LINE	1K DAU	50K DAU	500K DAU
X reseller stream + re-poll	\$1,603	\$9,671	\$27,300
TikTok / Instagram collection	\$50	\$3,750	\$16,400
Broadcast ASR	\$190	\$570	\$2,300
Core compute	\$420	\$2,400	\$12,000
Redpanda	\$50	\$450	\$1,200
Postgres + pgvector + Timescale	\$240	\$1,815	\$4,400
Dragonfly	\$95	\$390	\$2,340
ClickHouse	\$200	\$900	\$3,500
Cohere embed-v4	\$1	\$4	\$16
LLM namer + safety	\$106	\$301	\$904
OCR, NER, CLIP	\$106	\$468	\$2,173
Solana RPC	\$150	\$2,800	\$13,800
Execution cells (bare metal)	\$0	\$930	\$4,600
Provenance anchoring	\$60	\$64	\$75

THE \$42,000/MONTH COIN-FLIP

Ingestion forces X Enterprise at 500k DAU by counting **post-days** (7.6M reads/month, blowing the 2M pay-per-use cap). Backend says Enterprise is never triggered by counting **distinct posts** — a **25× discrepancy** and a \$42,000/month swing that nobody had noticed. Post-day accounting is the only arithmetically consistent one; route hot re-polls to the reseller and hold Enterprise in reserve.

13

What the review found

Eight specialists designed independently. A principal-engineer pass then hunted for contradictions. It found twelve, plus seven responsibilities nobody claimed.

Contradictions

- 01 Four message buses, no bridge.** Ingestion and SRE build Redpanda. Detection and clustering consume NATS. Backend rejects both for Postgres logical replication. The missing bridge is *on the critical path* and appears in no cost model.
- 02 Three key spaces on the same objects.** `post_id` is a namespaced string, a `u128`, and a `uuid` depending on the layer. Any join across them is broken.
- 03 Age decay specified three ways, disagreeing 12× at the floor.**
 `$0.6 + 0.4 * 2^{(-t/360)}$` VS `$0.05 + 0.95 * 2^{(-t/45)}$` VS `$2^{(-t/45)}$` with no floor.
- 04 The out-of-network inversion has four formulas spanning 8× at 100 followers** — and the Sybil pricing is calibrated against the weakest one.
- 05 Poll cadence off by 6×** — 60s vs 10s — and the hot-lane cost model rests on the higher figure.
- 06 UTC-day dedupe accounting differs 25×**, swinging \$42,000/month.
- 07 Protocol mismatch at the client boundary.** Backend serves SSE over HTTP/2 with `Last-Event-ID` gap-fill; the client consumes WebSocket MessagePack.

- 08 **TxGuard hard-blocks every Phase 1 launch** and has no override path by design, because it decodes the split at an offset in a program that will not exist yet.
- 09 **IP risk blocks launches in two layers and is forbidden from blocking in two others** — on the exact Chill Guy scenario the feature exists for.
- 10 **Coin image policy is mutually exclusive** — source crop vs derived render — and *both* sides advance their choice as the legal defence.
- 11 **Lead-time t_0 has two contradictory definitions**, and one of them is inflatable in the flattering direction. Use both endpoints on our own clock so measurement error cancels rather than accumulates.
- 12 **Revenue disagrees 10x on creator fee** (1% vs 0.1%) — which flips the seed tier between viable and dead.

Unowned responsibilities

- ! **Story-to-existing-coin matching** — the entire buy side, and the majority of trade volume.
- ! **The Holding object** — nobody owns reconciling on-chain balances with position rows, or cost basis and PnL.
- ! **The Follow object** — appears twice in eight designs, owned by none.
- ! **The creator 50% and treasury 10% claim paths.** The poster's 40% leg is designed in extreme depth; the other two legs are not designed at all.
- ! **Broadcast ingestion** — three layers budget ASR for it, zero build an adapter.
- ! **The Pro-mode query language** — half the declared UI surface, referenced by three layers, specified by none.

Build plan

PHASE	WEEKS	SHIPS	PROVES
0 Kill assumptions	1–3	Eight cheap experiments, no code. The \$0.02 dedupe test. Bags handle resolution for a never-onboarded handle. The real fee schedule.	Whether the cost model survives contact. Three of these can each independently invalidate the plan.
1 One bus, one clock, one ledger	4–11	Redpanda as the single bus with a schema registry. One id scheme (uuidv7 + platform natural key). Postgres 18 + pgvector. The provenance ledger and hourly anchor.	That we can prove we knew something at a time, to a sceptic, with no dependency on us. The only component that cannot be rented.
2 Detection in shadow	8–18	HOT scoring with one reconciled Omega and one age decay. Clustering with the reconciler and merge events. Nothing publishes.	That the lead time is real and measurable without flattering ourselves.
3 Read-only product	16–24	Public anonymous feed, Simple mode, share cards rendered from signed lead receipts.	That the share loop works and the number survives public scrutiny. The cheapest way to find out if anyone cares.
4 Trading	22–32	Privy wallets behind one WalletPort. Adapter-aware TxGuard. Jupiter routing. Story-to-coin matching.	That revenue exists and the buy side works. Trading ships <i>before</i> creation deliberately — lower risk, no IP exposure.
5 Creation and the 40% promise	28–40	The auto-namer with all four default-deny safety gates. The opt-out registry wired into all enforcement points. PosterVault.	That the 50/40/10 promise is honourable end to end, including for a poster who never asked to participate.
6 Sovereignty and Pro	36–52	Own fee router on raw Meteora DBC, removing the Bags dependency. Two independent audits. Pro mode.	That the company survives losing any single vendor — the launchpad, the reseller, or X itself.

Legal and compliance

Right of publicity

Routing fees to a creator's handle without consent is state-law governed with no federal statute, and **consent is the primary defence** [23]. Mitigations must ship in the *same release* as the split, never after: a permanent opt-out registry, a "not created or endorsed by" disclaimer on every origin coin, and per-source rate limits so one viral post cannot spawn 200 dilutive launches.

The Chill Guy precedent

Phillip Banks announced DMCA action against unauthorised merchandise and "shitcoins" [22]; the token fell roughly **54% in two hours**. Enforcement stalled because the deployer was anonymous and the token permissionless. The lesson is not that takedowns work — it is that **IP risk is a price-moving, tradeable signal**, which is why it is a first-class column in Pro mode.

Front-end registration exposure

SEC staff issued broker-dealer registration guidance for certain user interfaces in April 2026 [21]. A fee-charging front end that suggests an asset and offers one-click deployment is the exact profile that undermines a neutral-software defence. Concrete mitigations: present three suggestions and never one recommendation; never rank by predicted price performance; keep every transaction user-signed; retain no upgrade authority; route through a third-party launchpad rather than operating a curve.

Scraping posture

A federal court held that the Copyright Act preempts X's web-scraping claims [24], which is favourable on one axis. But the risk that kills a live-feed product is not a lawsuit — it is a Tuesday morning when the feed goes dark. The reseller sits in the *hiQ* posture (logged-in, contractually liable), not the *Meta v. Bright Data* safe harbour. Keep the licensed lane running even when it is not the cheapest.

16

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24 primary sources. 120 claims extracted, 25 adversarially verified, 16 confirmed, **9 killed**.

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CLAIMS THAT VERIFICATION KILLED — DO NOT BUILD ON THESE

The 2026 Phoenix 15-action probability set · Kleinberg's cost recurrence being "identical in form to Viterbi" · S-H-ESD being explicitly designed for local anomalies · the Bags social-provider list covering all ingested platforms · fee splits summing to exactly 10,000 bps · fee-share config being immutable after launch. Several of these were asserted earlier in this project and are now retracted.

Insidor engineering architecture v1.0 · generated 22 July 2026

Two research passes: adversarial verification of 120 claims against 24 primary sources,
and an eight-discipline design reviewed for cross-layer contradictions.

Figures marked UNVERIFIED or ASSUMPTION are exactly that. Phase 0 exists to kill them.